

# **Semester Project**

## **(Sports Car Showroom Management System)**



### **Submitted by:**

|                |             |
|----------------|-------------|
| Wajeeha Javed  | 2024-CS-601 |
| Minahil Khalid | 2024-CS-604 |
| Rohma Shahid   | 2024-CS-631 |

### **Submitted to:**

Mr. Zeeshan Ramzan

**Department of Computer Science**  
**University of Engineering and Technology**  
**New Campus**

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# Main Report Section

## 1. Abstract

The **Sports Car Showroom Management System** is a web-based application developed using **ASP.NET Core MVC** and **C#**, designed to automate and digitalize the management of a car showroom. The system provides a smooth interface for both customers and administrators. Customers can browse available sports cars, view details such as model, price, and specifications, and simulate purchases. Administrators can log in to manage the inventory by adding or deleting car records. Data handling is implemented using **database**, ensuring quick and simple management during development. This project reduces manual work, streamlines the showroom's operations, and demonstrates how modern web technologies can create efficient, user-friendly management systems.

## 2. Introduction

Managing car showrooms manually often leads to inefficiency, data redundancy, and slow updates. To overcome these challenges, the **Sports Car Showroom Management System** has been developed using **ASP.NET Core MVC** and **C#**. This system provides a structured and interactive platform that connects customers and administrators under one digital environment.

The application consists of two main modules:

- **Customer Module** – allows users to register, log in, view available sports cars, check detailed specifications, and perform simulated purchases.
- **Admin Module** – enables administrators to log in securely, add new cars, delete existing records, and manage the displayed inventory.

The system is implemented using **database** for simplicity, making it efficient for small-scale or demo environments. It focuses on providing a responsive and intuitive user interface with core functionalities of car browsing and management. Overall, this project represents a practical implementation of

ASP.NET MVC concepts while offering a real-world solution for showroom digitalization.

### **3. Vision**

To digitalize the sports car showroom experience by creating an efficient, user-friendly web platform that simplifies car management for admin and provides an engaging browsing experience for customers.

### **4. Problem Statement**

How can we overcome the challenges faced by traditional car showroom management systems to create a more efficient, secure, and user-friendly digital platform for both administrators and customers to manage and explore sports cars effectively?

### **5. Objectives**

- **User-Friendly Interface:**  
To design a responsive and visually appealing interface that allows both administrators and customers to easily navigate the system and access relevant functionalities without technical difficulty.
- **Automated Car Inventory Management:**  
To implement a system that allows the admin to add, update, and delete car records efficiently, minimizing manual work and reducing errors in car listings.
- **Customer Account Management:**  
To allow customers to register, log in, and enabling them to view car details and simulate purchase actions easily.
- **Car Purchase Simulation:**  
To enable customers to view car specifications ,prices, and simulate the buying process in a simple and interactive way.
- **Secure Login System:**  
To implement role-based access control, ensuring secure login for both customers and administrators using authentication in ASP.NET Core MVC.
- **Database Integration:**  
To use a structured database for data storage and retrieval, ensuring quick system response and smooth functionality during development and testing.
- **Admin Control Panel:**  
To provide administrators with full control over car listings, allowing them to update information, delete cars , and keep the online catalog up to date.

## 6. Scope

The scope of the **Sports Car Showroom Management System** covers the core functionalities for both administrators and customers. Administrators can securely log in, manage the car inventory by adding, updating or deleting car records, and ensure that the displayed information such as model, price, and specifications remains updated. Customers can register, log in, browse the available sports cars, view detailed information about each vehicle, and simulate the process of buying a car. The system focuses on creating a simple, fast, and interactive web platform developed in **ASP.NET Core MVC** using **C#**, with data stored in a **database** for ease of management and testing. Overall, the system aims to automate showroom operations, improve accessibility, and provide an efficient digital experience for both customers and administrators.

## 7. Tools and Technology

- **Programming Language:** C#
- **Documentation:** WPS Word
- **Framework:** ASP.NET Core MVC
- **Integrated Development Environment (IDE):** Visual Studio 2022
- **Mendeley:** References
- **Star UML:** For Diagrams
- **Figma:** Wireframes, Story Board

## 8. System Design

### I.System Flow (Logic)

- A customer can register and log in to view car listings.
- Admins can log in, add, or delete car entries from the system.
- The database updates dynamically when a purchase or car modification is made.
- Validation ensures that user credentials and car details are verified before saving

### II.Database design

The database consists of key entities such as:

- **Users Table:** Stores admin and customer credentials and profile details.
- **Cars Table:** Contains information about each sports car, including model, price, and availability.

## 9. Related Work

A number of studies have addressed digital solutions for car showroom and dealer management. Singh et al. (2021) developed a web-based *Car and Bike Showroom Management System* that allows customers to buy or book vehicles online. Similarly, the work by Beken et al. (2024) presents a *Dealer Management System for the Automotive Industry* with comprehensive inventory and customer-module coverage. In these contexts, our project builds on prior research by focusing specifically on **sports car showrooms**, leveraging **ASP.NET Core MVC** and **C#** with a **database backend** to improve administrative control and enhance customer interaction through a streamlined, web-based interface.

## 10. Team Members & Individual Tasks

| Name           | Registration No. | Individual Task                                                                                                       |
|----------------|------------------|-----------------------------------------------------------------------------------------------------------------------|
| Wajeetha Javed | 2024-CS-601      | <ul style="list-style-type: none"><li>● User stories</li><li>● Use cases</li></ul>                                    |
| Minahil Khalid | 2024-CS-604      | <ul style="list-style-type: none"><li>● Vision</li><li>● Abstraction</li><li>● Introduction</li><li>● Scope</li></ul> |
| Rohma Shahid   | 2024-CS-631      | <ul style="list-style-type: none"><li>● Functional Requirements</li><li>● Wireframes</li><li>● Story board</li></ul>  |

## 11. Timeline

| Phase  | Description                                                   | Duration | Team Member Involved |
|--------|---------------------------------------------------------------|----------|----------------------|
| Week 1 | Project idea selection, defining project scope and objectives | 1 week   | All Member           |
| Week 2 | Literature review and related work research                   | 1 week   | Wajeetha Javed       |
| Week 3 | System design                                                 | 1 week   | Minahil Khalid       |
| Week 4 | wireframe creation                                            | 1 week   | Rohma Shahid         |

## **12. Data Gathering Approach**

The data for this system was gathered by manually entering car details such as name, model, price, company, and specifications through the admin panel. The admin is responsible for updating, adding, or deleting car information as needed. This approach ensures that the displayed data is accurate and up to date.

# Requirements section

## 1.1 Functional Requirments

### Admin

Table 1 : Admin

| Index    | Requirments                                                                                                           |
|----------|-----------------------------------------------------------------------------------------------------------------------|
| FR-01-01 | The system shall allow an admin to <b>login</b> by providing an email , password , and role type .                    |
| FR-01-02 | The system shall verify admin credentials and allow access to the admin panel upon successful authentication.         |
| FR-01-03 | The system shall allow the admin to view all cars currently available in the showroom.                                |
| FR-01-04 | The system shall enable the admin to add car details(such as car name , model , price , and description ) to the list |
| FR-01-05 | The system shall allow the admin to delete any car from the showroom by entering its unique ID                        |
| FR-01-06 | The system shall allow the admin to update any car from the showroom by entering its unique ID                        |
| FR-01-07 | The system shall allow the admin to logout sucessfully                                                                |

### Customer

Table 2 : Customer

| index    | Requirments                                                                                                                                     |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-02-01 | The system shall allow customers to create an account by entering their name, email , and password, so they can access the showroom's features. |
| FR-02-02 | The system shall authenticate customer credentials and grant access to the system when valid information is provided.                           |
| FR-02-03 | The system shall display a complete list of sports cars available in the showroom , showing each car's name, model, and image.                  |

|          |                                                                                                                                                    |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-02-04 | The system shall display detailed information about a selected car including price , specifications, and description when user clicks on “details” |
| FR-02-05 | The system shall allow the customer to simulate the purchase of a car and show a confirmation message after successful buying                      |

## **1.2 Non-functional Requirments**

### **1. User Registration:**

Customers shall be able to create an account by entering valid personal information into a registration form.

### **2. User Authentication:**

Registered customers shall authenticate via the login page using their credentials before accessing user-specific functionalities.

### **3. Browse Car Listings:**

Authenticated and guest customers shall be able to view a catalog of available cars presented with brief details and images.

### **4. View Car Details:**

Customers shall be able to click on a specific car to access a detailed view containing complete specifications and pricing information.

### **5. Simulated Purchase:**

Authenticated customers shall initiate a simulated purchase using the “Buy Now” feature, triggering a confirmation message upon completion.

### **6. Purchase Confirmation:**

After a simulated transaction, the system shall display a confirmation message summarizing the car details and simulated purchase outcome.

## **7. Logout Functionality:**

Customers shall be able to securely log out at any time, returning to the public browsing interface without session data persistence

## **1.3 Business Requirements**

- **Centralized Car Information Management**

The system must allow the admin to add, update, and delete details of sports cars, including model name, brand, price, and specifications. This helps the dealership maintain accurate and organized records.

- **Car Display for Users**

The system must display all available sports cars in a visually appealing interface where users can view details, specifications, and images of each car.

- **Admin Control Panel**

The system must include an admin panel where the administrator can manage car data without needing technical expertise.

- **Data Security and Accuracy**

The system must ensure that data entered by the admin is stored correctly in the database and displayed accurately on the website.

- **Responsive and User-Friendly Interface**

The system must provide a simple, attractive, and responsive web interface that adjusts well to different devices and screen sizes.

- **Maintainability and Scalability:**

The system must be easy to update for future improvements such as adding customer accounts, analytics, or maintenance tracking modules.

## **1.4 Bussiness Rules**

1. Only registered users can make purchases or place orders.
2. The admin is the only authorized user who can add, update, or delete car information.
3. A valid email and password are required for login authentication.
4. All cars added by the admin must include mandatory details (name, company, model, and price).
5. Once an order is confirmed, it cannot be modified or deleted by the user.
6. System should ensure data consistency. Any car deleted by the admin should automatically be removed from the user's view.

## **1.5 User Requirments**

### **Customer:**

- View the list of sports cars with basic details and images.
- View complete details of any car.
- Register and log in to their account securely.
- Buy cars using the “Buy Now” option.
- Receive confirmation after successful order placement.

### **Admin:**

- Log in to the admin panel.
- Add, edit, or delete car information.
- View all car listings.
- Manage user details if required.
- Log out securely after completing tasks.

## **1.6 External Interface**

### **User Interface:**

Developed using HTML, CSS, and Razor views in ASP.NET Core MVC. It provides forms, buttons, and menus for easy navigation

### **Database Interface:**

The system interacts with database to store and retrieve car, admin, and user data.

## **1.7 Physical Requirements**

- Laptop
- Power Supply
- Internet
- 4 GB RAM and Core i3 Processor

## **1.7 Development Constraints**

- The system is designed for local database use, not a live online server.
- Developed using ASP.NET Core MVC (C#) framework —limited to features supported by Visual Studio.
- Testing is performed locally; no third-party API integration.
- Limited time duration restricted implementation of advanced modules such as payment gateways or car test drives.

# Design Specifications

## 2.1 Wireframes

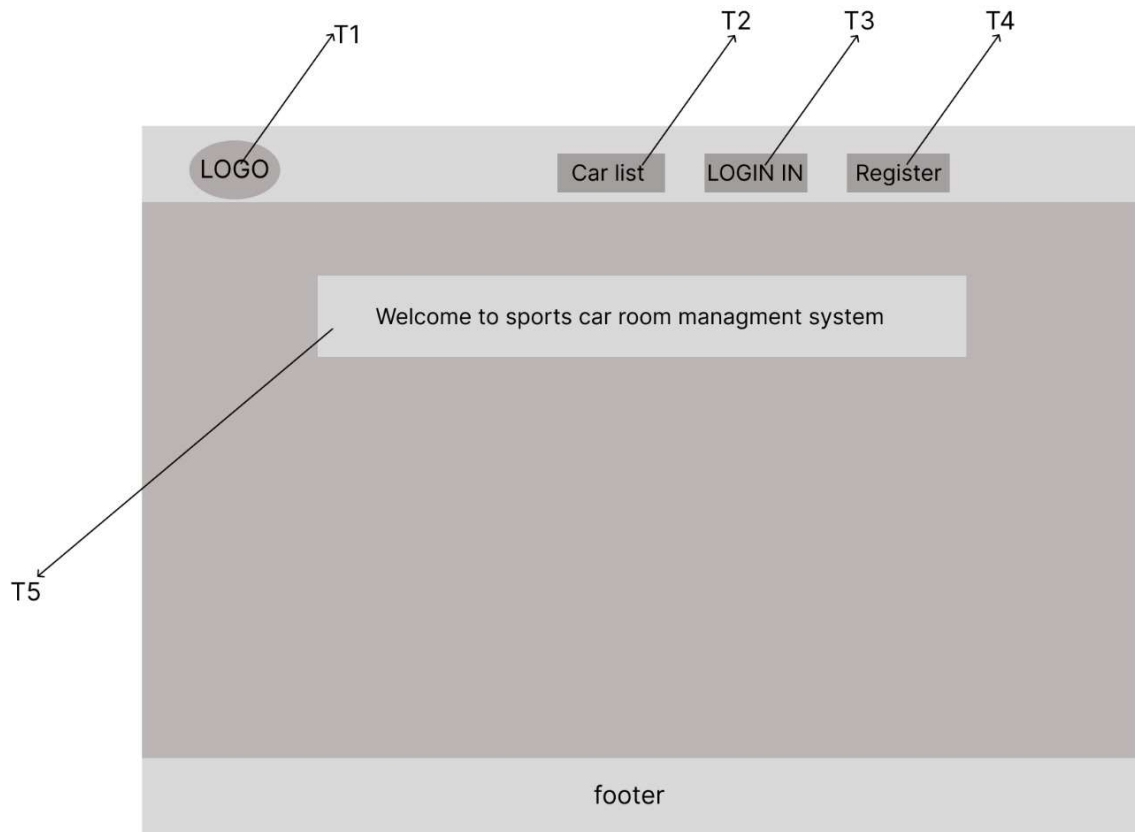


Figure 1: Home wire frame [1]

| Sr NO. | Element       | As Customer                     | As Admin                        |
|--------|---------------|---------------------------------|---------------------------------|
| T1     | Logo          | I shall jump to the home page   | I shall jump to the home page   |
| T2     | Car list      | I shall view the car list       | I shall view the car list       |
| T3     | Login         | I shall go to login page        | I shall go to login page        |
| T4     | Register      | I shall go to registration page | I shall go to registration page |
| T5     | Welcome title | I shall view welcome page       | I shall view welcome page       |

Table 3: Home

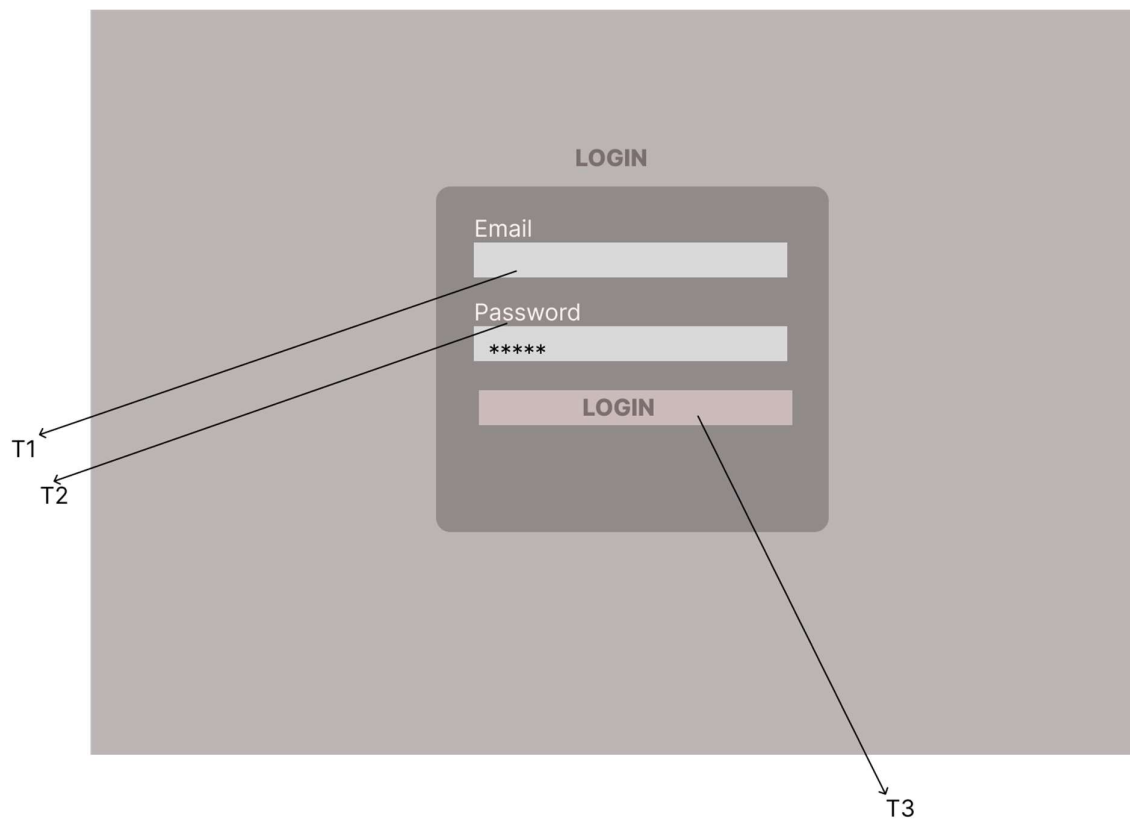


Figure 2:Login wire frame [2]

| Sr NO. | Element      | As Customer                         | As Admin                           |
|--------|--------------|-------------------------------------|------------------------------------|
| T1     | Email        | Input field for customer's email    | Input field for admin's email      |
| T2     | Password     | Input field for customer's password | Input field for admin's password   |
| T3     | Login button | Button to log in as a customer      | Admin will jump to Admin dashboard |

Table 4: Login

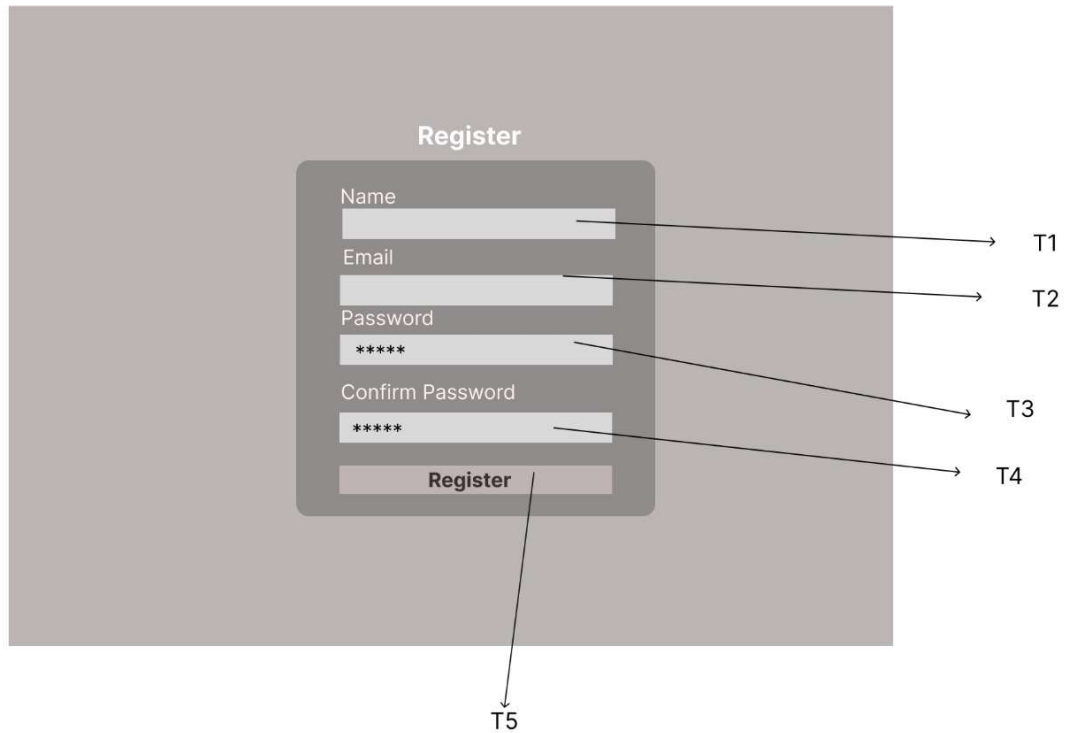


Figure 3: Register wire frame [3]

| Sr NO. | Element            | As Customer                                   | As Admin       |
|--------|--------------------|-----------------------------------------------|----------------|
| T1     | Name               | Input field for customer's name               | Not Applicable |
| T2     | Email              | Input field for customer's email              | Not Applicable |
| T3     | Password           | Input field for customer's password           | Not Applicable |
| T4     | Confirmed Password | Input field for customer to re-enter password | Not Applicable |
| T5     | Register Button    | By clicking this , I register as customer     | Not Applicable |

Table 5: Register

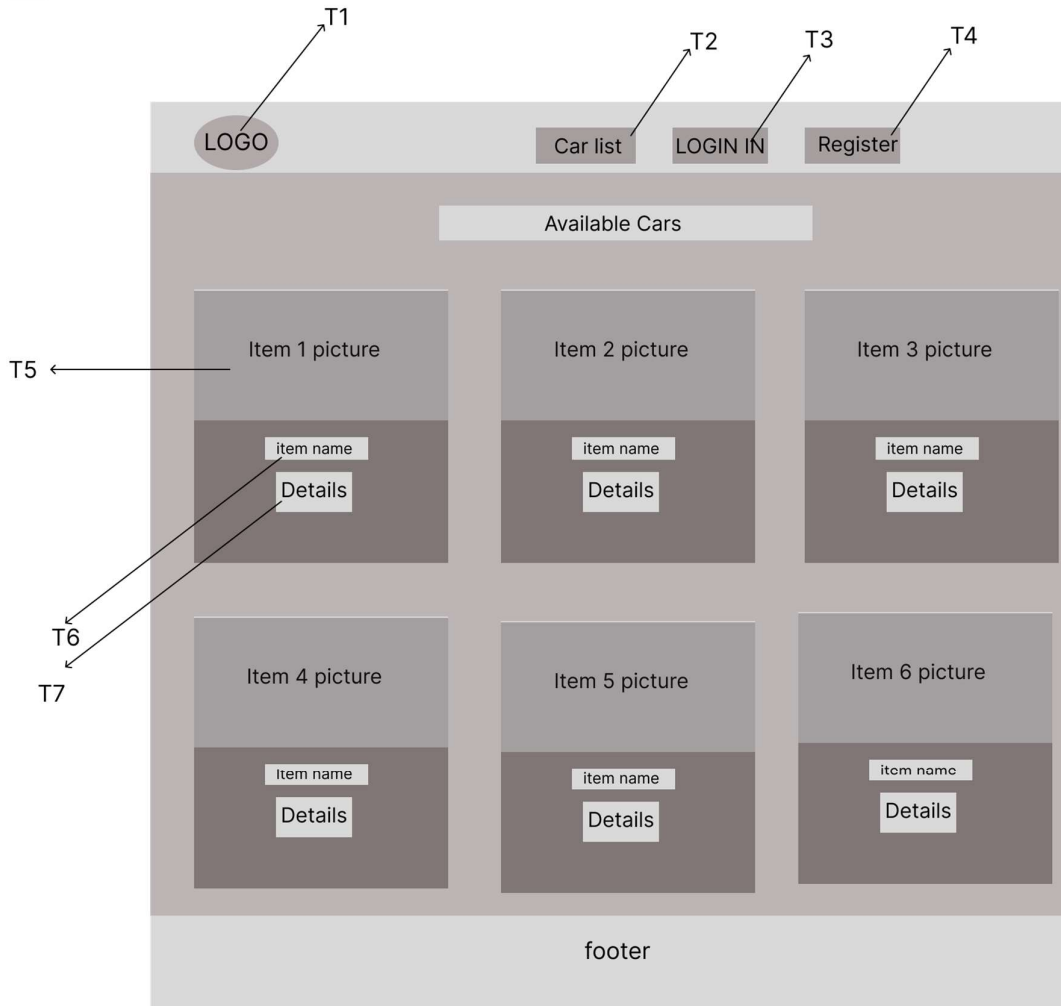


Figure 4:car\_list wire frame [4]

| Sr NO. | Element    | As Customer                     | As Admin                      |
|--------|------------|---------------------------------|-------------------------------|
| T1     | Logo       | I shall jump to the home page   | I shall jump to the home page |
| T2     | Car list   | I shall view the car list       | I shall view the car list     |
| T3     | Login      | I shall go to login page        | I shall go to login page      |
| T4     | Register   | I shall go to registration page | Not applicable                |
| T5     | Item image | I shall view the car image      | I shall view the car image    |
| T6     | Item name  | I shall view the car name       | I shall view the car name     |
| T7     | Details    | I shall jump to details page    | I shall jump to details page  |

Table 6: Car\_list

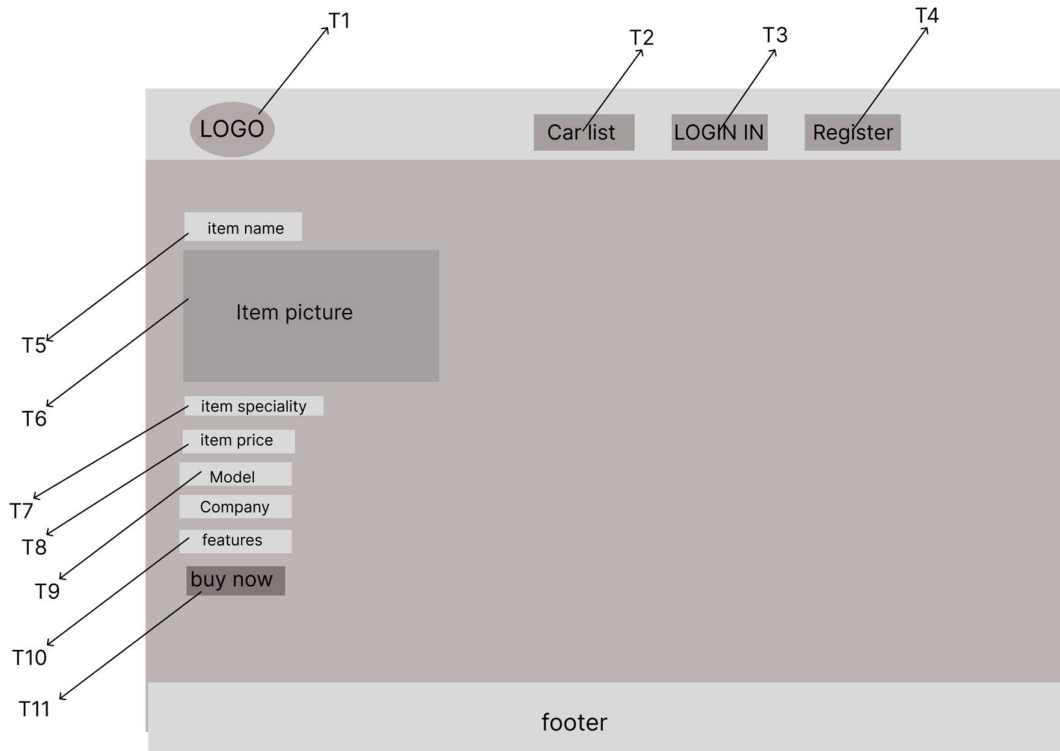


Figure 5:Car details wire frame [5]

| Sr NO. | Element         | As Customer                     | As Admin                      |
|--------|-----------------|---------------------------------|-------------------------------|
| T1     | Logo            | I shall jump to the home page   | I shall jump to the home page |
| T2     | Car list        | I shall view the car list       | I shall view the car list     |
| T3     | Login           | I shall go to login page        | I shall go to login page      |
| T4     | Register        | I shall go to registration page | Not Applicable                |
| T5     | Item name       | I shall view car name           | I shall view car name         |
| T6     | Item image      | I shall view car image          | I shall view car image        |
| T7     | Item speciality | I shall view car speciality     | I shall view car speciality   |
| T8     | Item price      | I shall view car price          | I shall view car price        |
| T9     | Model           | I shall view car model          | I shall view car model        |
| T10    | Features        | I shall view car features.      | I shall view car features.    |
| T11    | Buy button      | I shall buy car                 | Not applicable                |

Table 7: Car details

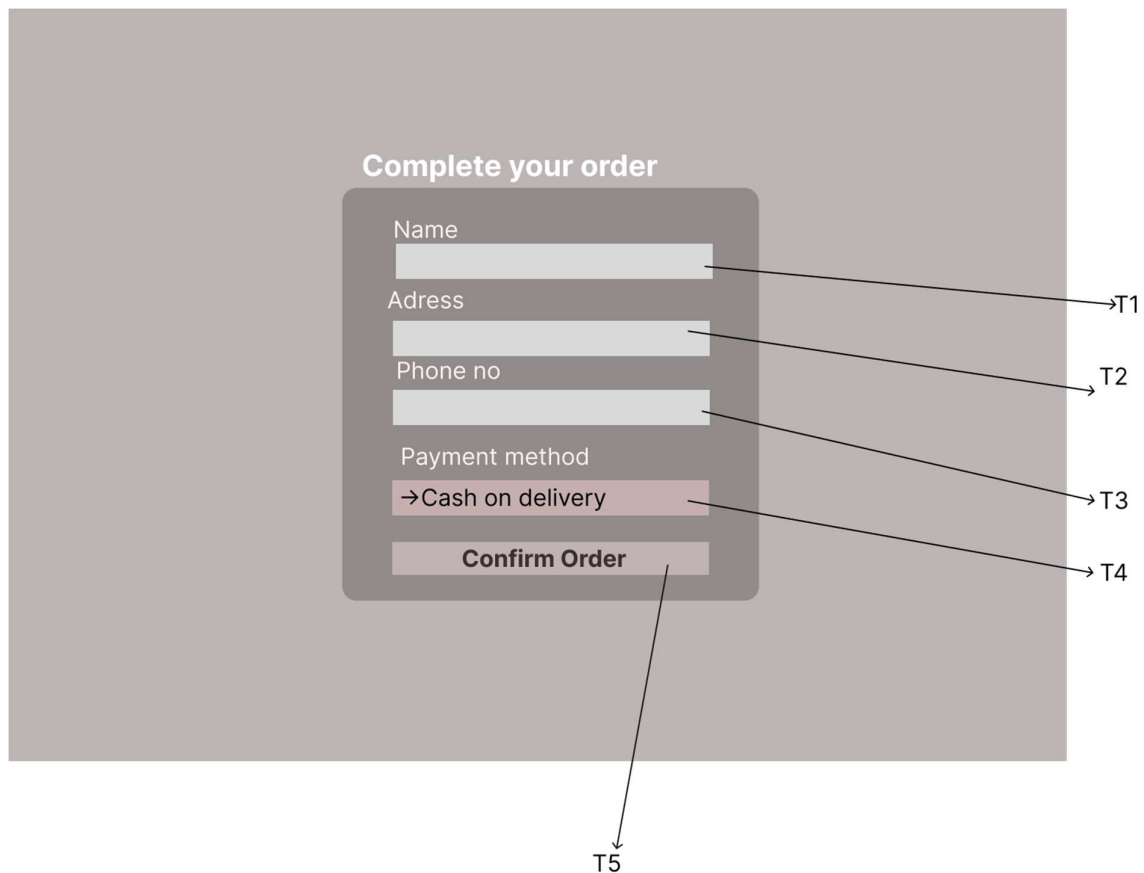


Figure 6: Order Details wireframe[6]

Table 8: Order Details

| Sr NO. | Element              | As Customer                                              | As Admin       |
|--------|----------------------|----------------------------------------------------------|----------------|
| T1     | Name                 | I shall enter my name in input field                     | Not Applicable |
| T2     | Address              | I shall enter my email in input field                    | Not Applicable |
| T3     | Phone No.            | I shall enter my phone number in input field             | Not Applicable |
| T4     | Payment Method       | I shall select Cash On Delivery as payment method        | Not Applicable |
| T5     | Confirm Order Button | I must click "confirm Order Button" to complete my order | Not Applicable |

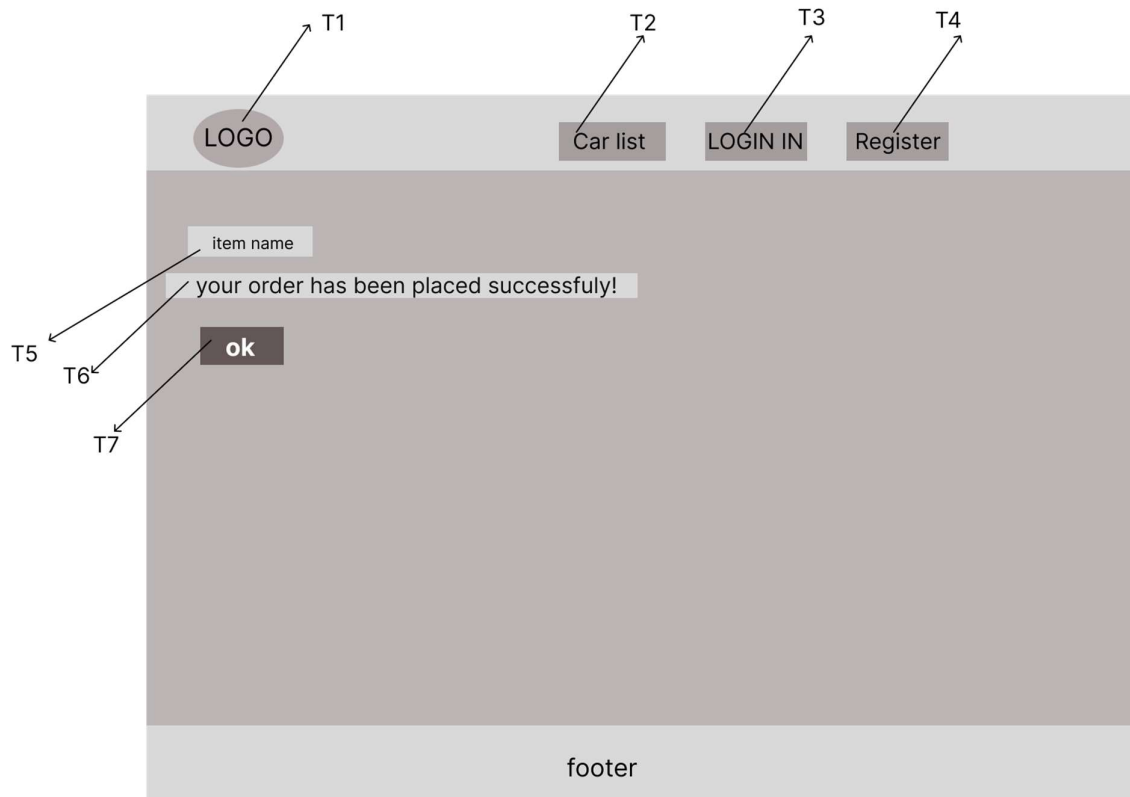


Figure 7: Buy confirmation wire frame [6]

| Sr NO. | Element              | As Customer                                       | As Admin                      |
|--------|----------------------|---------------------------------------------------|-------------------------------|
| T1     | Logo                 | I shall jump to the home page                     | I shall jump to the home page |
| T2     | Car list             | I shall view the car list                         | I shall view the car list     |
| T3     | Login                | I shall go to login page                          | I shall go to login page      |
| T4     | Register             | I shall go to registration page                   | Not applicable                |
| T5     | Item name            | I shall view car name                             | Not applicable                |
| T6     | Confirmation message | I shall know that my order is placed successfully | Not applicable                |
| T7     | Ok Button            | I shall click on OK button to return to home page | Not applicable                |

Table 9: Buy confirmation

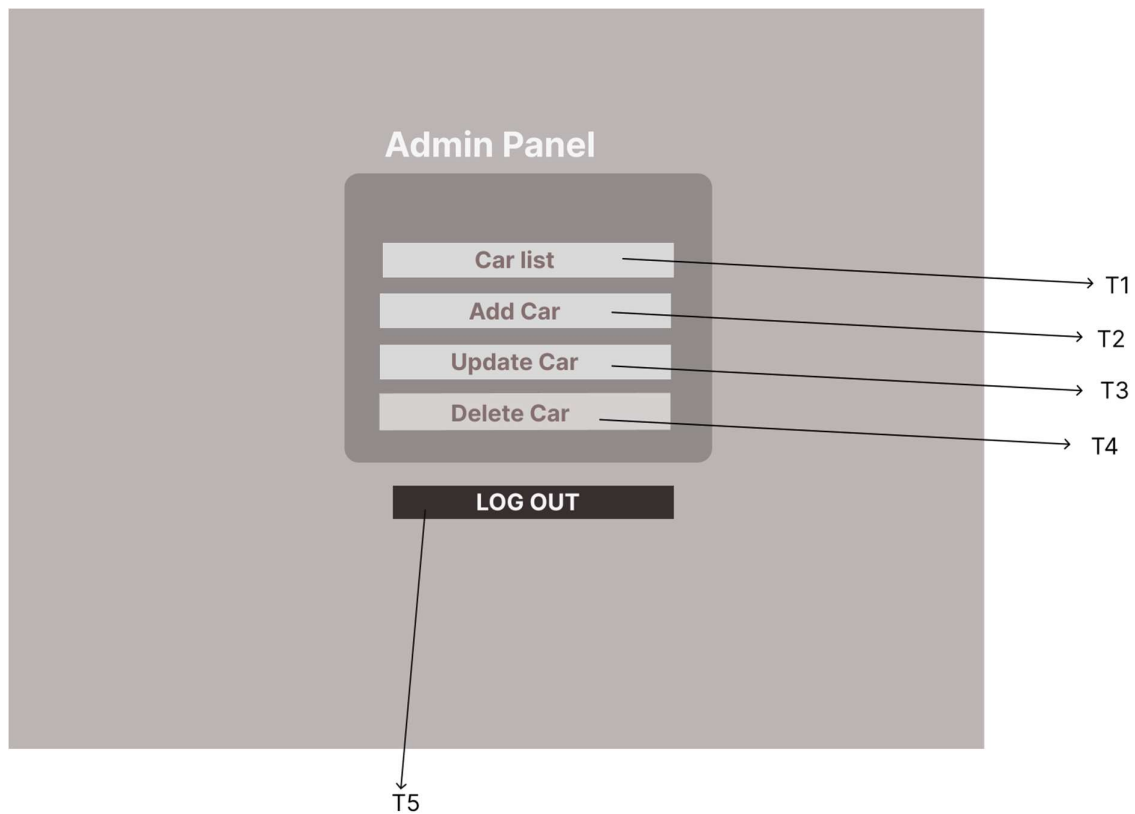


Figure 8:Admin Panel wire frame [7]

Table 10 : Admin Panel

| Sr NO. | Element    | As Admin                                 |
|--------|------------|------------------------------------------|
| T1     | Car List   | I shall view list of available cars      |
| T2     | Add Car    | I shall add new car in car list          |
| T3     | Update Car | I shall update details of available cars |
| T4     | Delete Car | I shall delete car from the car list     |
| T5     | Log Out    | I shall log out from Admin dashboard     |

The image shows a web form titled "Add a new car". The form contains four input fields: "Car Name", "Description", "Image Url", and "Price". Below these fields is a dark button labeled "Add Car". Five arrows point from labels T1 through T5 to the form elements: T1 points to the "Car Name" field, T2 points to the "Description" field, T3 points to the "Image Url" field, T4 points to the "Price" field, and T5 points to the "Add Car" button.

Figure 9 :Add New Car

Table 11:Add New Car

| Sr NO. | Element         | As Admin                                     |
|--------|-----------------|----------------------------------------------|
| T1     | Car Name        | I shall enter car name in input field        |
| T2     | Car Description | I shall enter car description in input field |
| T3     | Car image       | I shall upload car image                     |
| T4     | Car Price       | I shall enter car price                      |
| T5     | Add Car button  | I shall add car by clicking button           |

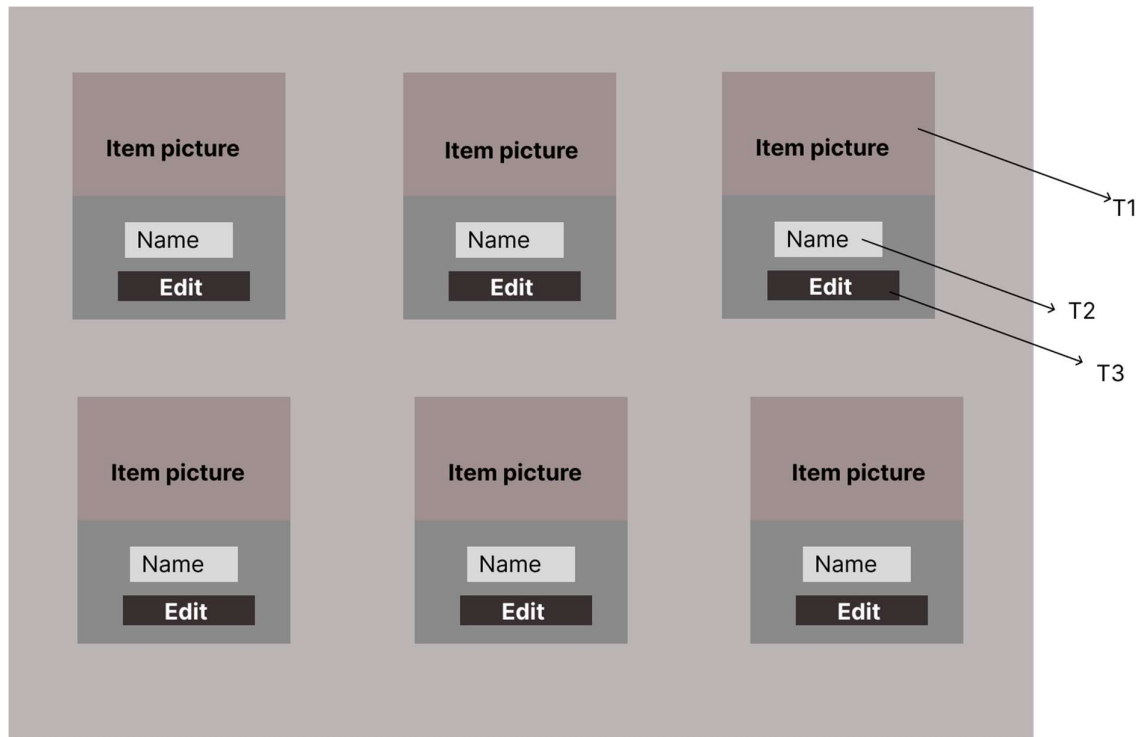


Figure 10 :Edit Car List

Table 12 :Edit Car List

| Sr NO. | Element     | As Admin                 |
|--------|-------------|--------------------------|
| T1     | Car Image   | I shall see car image    |
| T2     | Car Name    | I shall see car name     |
| T3     | Edit Button | I shall edit car details |

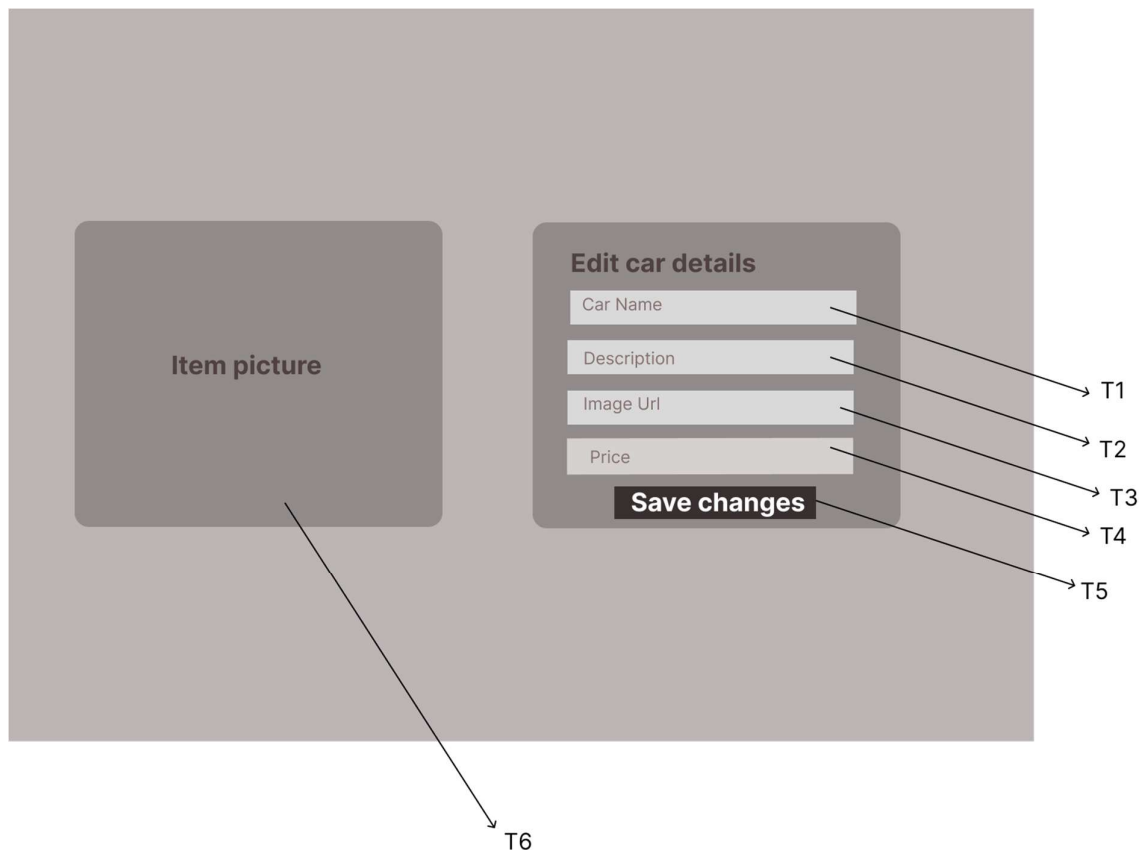


Figure 11: Edit Car

Table 13 :Edit Car

| Sr NO. | Element             | As Admin                     |
|--------|---------------------|------------------------------|
| T1     | Car Name            | I shall edit car name        |
| T2     | Car Description     | I shall edit car description |
| T3     | Car Image           | I shall edit car image       |
| T4     | Car Price           | I shall edit car price       |
| T5     | Save Changes Button | I shall save all changes     |
| T6     | Car image preview   | I shall see car image        |

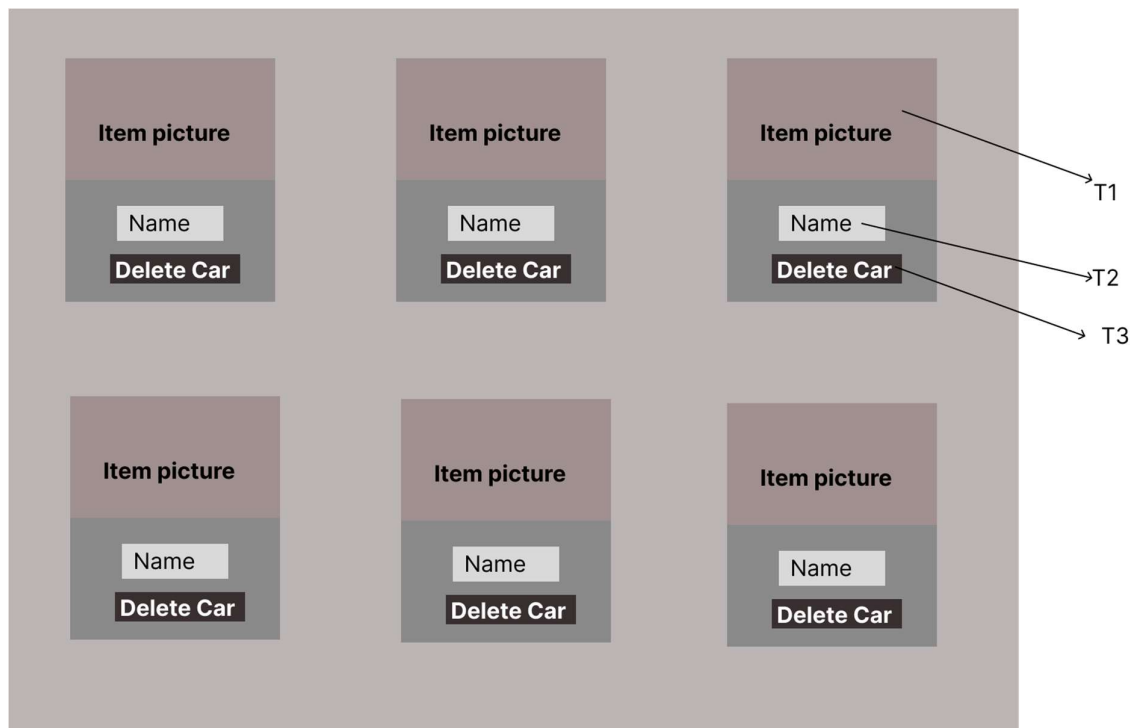


Figure 12 : Delete Car List

Table 14 :Delete Car

| Sr NO. | Element       | As Admin              |
|--------|---------------|-----------------------|
| T1     | Car Image     | I shall see car image |
| T2     | Car Name      | I shall see car name  |
| T3     | Delete Button | I shall delete car    |

## 2.2 Use cases

### Admin Login

Table 15 :Admin login use case

|                     |                                                                         |                                                                              |
|---------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Name                | Adminlogin                                                              |                                                                              |
| Participating Actor | Admin                                                                   |                                                                              |
| Goals               | To access the admin panel for managing car data.                        |                                                                              |
| Triggers            | When the admin clicks on the “Login” button                             |                                                                              |
| Pre-conditions      | i. Admin account must exist<br>ii. Credentials must be valid            |                                                                              |
| Post-conditions     | Admin dashboard opens                                                   |                                                                              |
| Basic Flow          | Actor                                                                   | System                                                                       |
|                     | 1. Admin clicks “Login”<br>2. Admin view dashboard                      | 1. The System verifies credentials<br>2. The System Redirects to admin panel |
| Alternative Flow    | Not available                                                           |                                                                              |
| Exceptions          | Invalid credentials<br>Solution: Show “incorrect username or password”. |                                                                              |
| Qualities           | ● Fast<br>● Secure<br>● responsive                                      |                                                                              |

## Add Car

Table 16 : Add car use case

|                                                              |                                                                                                                                                                                                                                           |  |       |        |                                                              |                                                                                             |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------|--------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Name                                                         | Add Car                                                                                                                                                                                                                                   |  |       |        |                                                              |                                                                                             |
| Participating Actor                                          | Admin                                                                                                                                                                                                                                     |  |       |        |                                                              |                                                                                             |
| Goals                                                        | To add new sports car into the system.                                                                                                                                                                                                    |  |       |        |                                                              |                                                                                             |
| Triggers                                                     | When the admin clicks on the “Add Car” button                                                                                                                                                                                             |  |       |        |                                                              |                                                                                             |
| Pre-conditions                                               | Admin must be logged in                                                                                                                                                                                                                   |  |       |        |                                                              |                                                                                             |
| Post-conditions                                              | New car record is added to the database                                                                                                                                                                                                   |  |       |        |                                                              |                                                                                             |
| Basic Flow                                                   | <table><tr><td>Actor</td><td>System</td></tr><tr><td>1.Admin clicks “Add Car”<br/>2.Admin enter details and submit</td><td>1.The System shows form to add details.<br/>2.The System stores car in database and confirms</td></tr></table> |  | Actor | System | 1.Admin clicks “Add Car”<br>2.Admin enter details and submit | 1.The System shows form to add details.<br>2.The System stores car in database and confirms |
| Actor                                                        | System                                                                                                                                                                                                                                    |  |       |        |                                                              |                                                                                             |
| 1.Admin clicks “Add Car”<br>2.Admin enter details and submit | 1.The System shows form to add details.<br>2.The System stores car in database and confirms                                                                                                                                               |  |       |        |                                                              |                                                                                             |
| Alternative Flow                                             | Not available                                                                                                                                                                                                                             |  |       |        |                                                              |                                                                                             |
| Exceptions                                                   | Missing required fields<br>Solution: Show “Please fill all details”.                                                                                                                                                                      |  |       |        |                                                              |                                                                                             |
| Qualities                                                    | <ul style="list-style-type: none"><li>● Fast</li><li>● User-friendly form</li><li>● Instant database</li></ul>                                                                                                                            |  |       |        |                                                              |                                                                                             |

## Edit Car

Table 17 : Editcar use case

|                                                        |                                                                                                                                                                                                                                    |  |       |        |                                                        |                                                                                            |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------|--------|--------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Name                                                   | Edit Car                                                                                                                                                                                                                           |  |       |        |                                                        |                                                                                            |
| Participating Actor                                    | Admin                                                                                                                                                                                                                              |  |       |        |                                                        |                                                                                            |
| Goals                                                  | To modify existing car details.                                                                                                                                                                                                    |  |       |        |                                                        |                                                                                            |
| Triggers                                               | When the admin clicks on the “Edit Car” button                                                                                                                                                                                     |  |       |        |                                                        |                                                                                            |
| Pre-conditions                                         | Car data must exist in database                                                                                                                                                                                                    |  |       |        |                                                        |                                                                                            |
| Post-conditions                                        | Car details are updated successfully.                                                                                                                                                                                              |  |       |        |                                                        |                                                                                            |
| Basic Flow                                             | <table><tr><td>Actor</td><td>System</td></tr><tr><td>1.Admin clicks “Edit Car”<br/>2.Admin modifies details.</td><td>1.The System shows form to edit details.<br/>2.The System updates car records in database .</td></tr></table> |  | Actor | System | 1.Admin clicks “Edit Car”<br>2.Admin modifies details. | 1.The System shows form to edit details.<br>2.The System updates car records in database . |
| Actor                                                  | System                                                                                                                                                                                                                             |  |       |        |                                                        |                                                                                            |
| 1.Admin clicks “Edit Car”<br>2.Admin modifies details. | 1.The System shows form to edit details.<br>2.The System updates car records in database .                                                                                                                                         |  |       |        |                                                        |                                                                                            |
| Alternative Flow                                       | Not available                                                                                                                                                                                                                      |  |       |        |                                                        |                                                                                            |
| Exceptions                                             | Invalid Input<br>Solution: Show “Please enter valid data”.                                                                                                                                                                         |  |       |        |                                                        |                                                                                            |
| Qualities                                              | <ul style="list-style-type: none"><li>● Smooth data update</li><li>● User-friendly form</li></ul>                                                                                                                                  |  |       |        |                                                        |                                                                                            |

## Delete Car

Table 18 : Delete car use case

|                                                 |                                                                                                                                                                                                                       |  |       |        |                                                 |                                                                                      |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------|--------|-------------------------------------------------|--------------------------------------------------------------------------------------|
| Name                                            | Delete Car                                                                                                                                                                                                            |  |       |        |                                                 |                                                                                      |
| Participating Actor                             | Admin                                                                                                                                                                                                                 |  |       |        |                                                 |                                                                                      |
| Goals                                           | To remove a car from the list.                                                                                                                                                                                        |  |       |        |                                                 |                                                                                      |
| Triggers                                        | When the admin clicks on the “Delete Car” button                                                                                                                                                                      |  |       |        |                                                 |                                                                                      |
| Pre-conditions                                  | Car record must exist.                                                                                                                                                                                                |  |       |        |                                                 |                                                                                      |
| Post-conditions                                 | Car is removed from database                                                                                                                                                                                          |  |       |        |                                                 |                                                                                      |
| Basic Flow                                      | <table><tr><td>Actor</td><td>System</td></tr><tr><td>1.Admin clicks “Delete Car”<br/>2.Admin confirms</td><td>1.The System confirms deletion.<br/>2.The System deletes car record and updates list.</td></tr></table> |  | Actor | System | 1.Admin clicks “Delete Car”<br>2.Admin confirms | 1.The System confirms deletion.<br>2.The System deletes car record and updates list. |
| Actor                                           | System                                                                                                                                                                                                                |  |       |        |                                                 |                                                                                      |
| 1.Admin clicks “Delete Car”<br>2.Admin confirms | 1.The System confirms deletion.<br>2.The System deletes car record and updates list.                                                                                                                                  |  |       |        |                                                 |                                                                                      |
| Alternative Flow                                | Not available                                                                                                                                                                                                         |  |       |        |                                                 |                                                                                      |
| Exceptions                                      | Car not found<br>Solution: Show “Record missing”.                                                                                                                                                                     |  |       |        |                                                 |                                                                                      |
| Qualities                                       | <ul style="list-style-type: none"><li>● Safe Deletion</li><li>● User-friendly form</li><li>● Instant database</li></ul>                                                                                               |  |       |        |                                                 |                                                                                      |

## User Login

Table 19 : System login use case

| Name                                                 | System login                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |       |        |                                             |                                                                    |                                                      |                                                |                             |                                                                     |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------|--------|---------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|-----------------------------|---------------------------------------------------------------------|
| Participating Actor                                  | Customer,Admin                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |       |        |                                             |                                                                    |                                                      |                                                |                             |                                                                     |
| Goals                                                | To allow the Customer or Admin to securely log into the system and access respective functionalities                                                                                                                                                                                                                                                                                                                                                |  |       |        |                                             |                                                                    |                                                      |                                                |                             |                                                                     |
| Triggers                                             | When the user opens the Sports Car Management System and clicks on the “Login” button                                                                                                                                                                                                                                                                                                                                                               |  |       |        |                                             |                                                                    |                                                      |                                                |                             |                                                                     |
| Pre-conditions                                       | iii. Customer or Admin must already have a registered account<br>iv. Internet connections must be available                                                                                                                                                                                                                                                                                                                                         |  |       |        |                                             |                                                                    |                                                      |                                                |                             |                                                                     |
| Post-conditions                                      | The Customer or Admin is successfully logged into their respective dashboard                                                                                                                                                                                                                                                                                                                                                                        |  |       |        |                                             |                                                                    |                                                      |                                                |                             |                                                                     |
| Basic Flow                                           | <table><tr><th>Actor</th><th>System</th></tr><tr><td>3. Customer/Admin enters email and password</td><td>3. The System receives and matches credentials from local storage.</td></tr><tr><td>4. Customer/Admin selects their role(Customer/Admin)</td><td>4. The System checks user role and credentials</td></tr><tr><td>5. Clicks on “Login” button</td><td>5. The System verifies details and redirect user to their dashboard</td></tr></table> |  | Actor | System | 3. Customer/Admin enters email and password | 3. The System receives and matches credentials from local storage. | 4. Customer/Admin selects their role(Customer/Admin) | 4. The System checks user role and credentials | 5. Clicks on “Login” button | 5. The System verifies details and redirect user to their dashboard |
| Actor                                                | System                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |       |        |                                             |                                                                    |                                                      |                                                |                             |                                                                     |
| 3. Customer/Admin enters email and password          | 3. The System receives and matches credentials from local storage.                                                                                                                                                                                                                                                                                                                                                                                  |  |       |        |                                             |                                                                    |                                                      |                                                |                             |                                                                     |
| 4. Customer/Admin selects their role(Customer/Admin) | 4. The System checks user role and credentials                                                                                                                                                                                                                                                                                                                                                                                                      |  |       |        |                                             |                                                                    |                                                      |                                                |                             |                                                                     |
| 5. Clicks on “Login” button                          | 5. The System verifies details and redirect user to their dashboard                                                                                                                                                                                                                                                                                                                                                                                 |  |       |        |                                             |                                                                    |                                                      |                                                |                             |                                                                     |
| Alternative Flow                                     | Not available                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |       |        |                                             |                                                                    |                                                      |                                                |                             |                                                                     |
| Exceptions                                           | <div>1. User doesn’t have an account .<br/>Solution:Must register first.</div> <div>2. Incorrect email entered.<br/>Solution: Enter correct email.</div> <div>3. Incorrect password entered<br/>Solution:Enter valid password.</div> <div>4. Wrong role selected.<br/>Solution:Choose correct user role.</div>                                                                                                                                      |  |       |        |                                             |                                                                    |                                                      |                                                |                             |                                                                     |
| Qualities                                            | <div>● Fast</div> <div>● Secure(local data)</div> <div>● responsive</div>                                                                                                                                                                                                                                                                                                                                                                           |  |       |        |                                             |                                                                    |                                                      |                                                |                             |                                                                     |

## Register Account

Table 20 : User Registration use case

|                                                          |                                                                                                                                                                                                          |  |       |        |                                                          |                                                                |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------|--------|----------------------------------------------------------|----------------------------------------------------------------|
| Name                                                     | User Registration                                                                                                                                                                                        |  |       |        |                                                          |                                                                |
| Participating Actor                                      | Customer                                                                                                                                                                                                 |  |       |        |                                                          |                                                                |
| Goals                                                    | To allow the Customer to create an account .                                                                                                                                                             |  |       |        |                                                          |                                                                |
| Triggers                                                 | When the user clicks on the “Register” button .                                                                                                                                                          |  |       |        |                                                          |                                                                |
| Pre-conditions                                           | User must not already exist                                                                                                                                                                              |  |       |        |                                                          |                                                                |
| Post-conditions                                          | User account is created .                                                                                                                                                                                |  |       |        |                                                          |                                                                |
| Basic Flow                                               | <table><tr><td>Actor</td><td>System</td></tr><tr><td>1. User clicks on “Register”.<br/>2. User enters details.</td><td>1. Displays registration form.<br/>2. Validates and stores user</td></tr></table> |  | Actor | System | 1. User clicks on “Register”.<br>2. User enters details. | 1. Displays registration form.<br>2. Validates and stores user |
| Actor                                                    | System                                                                                                                                                                                                   |  |       |        |                                                          |                                                                |
| 1. User clicks on “Register”.<br>2. User enters details. | 1. Displays registration form.<br>2. Validates and stores user                                                                                                                                           |  |       |        |                                                          |                                                                |
| Exceptions                                               | 1. Email already exists .<br>Solution:Use another email.<br>2. Missing details.<br>Solution: Fill all required fields                                                                                    |  |       |        |                                                          |                                                                |
| Qualities                                                | <ul style="list-style-type: none"><li>● User-friendly interface</li><li>● Data validation</li><li>● Account security</li></ul>                                                                           |  |       |        |                                                          |                                                                |

## View Car List

Table 21 :View Car List use case

|                     |                                                            |                                                                                    |
|---------------------|------------------------------------------------------------|------------------------------------------------------------------------------------|
| Name                | View Car List                                              |                                                                                    |
| Participating Actor | Customer,Admin                                             |                                                                                    |
| Goals               | To view the list of all available sports cars.             |                                                                                    |
| Triggers            | When the user clicks on the “Car List” option in the menu. |                                                                                    |
| Pre-conditions      | i. User must be logged in.<br>ii. Cars must be available . |                                                                                    |
| Post-conditions     | List of all cars is displayed with details.                |                                                                                    |
| Basic Flow          | Actor                                                      | System                                                                             |
|                     | 1. User clicks “Car List”.<br><br>2. User views cars.      | 1. System retrieves car data from local list<br>2. System displays data on webpage |
| Exceptions          | 1. Empty list .<br>Solution:Show “No cars available”.      |                                                                                    |
| Qualities           | ● User-friendly interface<br>● Quick display               |                                                                                    |

## View Car Details

Table 22 : View Car Details use case

| Name                                                | View Car Details                                                                                                                                                                                                            |  |       |        |                                                     |                                                                                        |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------|--------|-----------------------------------------------------|----------------------------------------------------------------------------------------|
| Participating Actor                                 | Customer,Admin                                                                                                                                                                                                              |  |       |        |                                                     |                                                                                        |
| Goals                                               | To show detailed information of a selected car                                                                                                                                                                              |  |       |        |                                                     |                                                                                        |
| Triggers                                            | When the user clicks on the “Detail” button .                                                                                                                                                                               |  |       |        |                                                     |                                                                                        |
| Pre-conditions                                      | Car details must be available in database .                                                                                                                                                                                 |  |       |        |                                                     |                                                                                        |
| Post-conditions                                     | Selected car’s full details shown.                                                                                                                                                                                          |  |       |        |                                                     |                                                                                        |
| Basic Flow                                          | <table><tr><th>Actor</th><th>System</th></tr><tr><td>1. User selects a car.<br/>2. User reads the details</td><td>1. System fetches that car’s info from database.<br/>2. System displays details of car.</td></tr></table> |  | Actor | System | 1. User selects a car.<br>2. User reads the details | 1. System fetches that car’s info from database.<br>2. System displays details of car. |
| Actor                                               | System                                                                                                                                                                                                                      |  |       |        |                                                     |                                                                                        |
| 1. User selects a car.<br>2. User reads the details | 1. System fetches that car’s info from database.<br>2. System displays details of car.                                                                                                                                      |  |       |        |                                                     |                                                                                        |
| Exceptions                                          | 1. Data missing .<br>Solution:Show “Details not available”.                                                                                                                                                                 |  |       |        |                                                     |                                                                                        |
| Qualities                                           | <ul style="list-style-type: none"><li>● Fast</li><li>● Reliable display</li><li>● Minimal delay</li></ul>                                                                                                                   |  |       |        |                                                     |                                                                                        |

# Buy Car

Table 23 : Buy Car use case

|                                                           |                                                                                                                                                                                                                                                          |  |       |        |                                                           |                                                                                                               |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------|--------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Name                                                      | Purchase Car                                                                                                                                                                                                                                             |  |       |        |                                                           |                                                                                                               |
| Participating Actor                                       | Customer                                                                                                                                                                                                                                                 |  |       |        |                                                           |                                                                                                               |
| Goals                                                     | To simulate buying a car from the system.                                                                                                                                                                                                                |  |       |        |                                                           |                                                                                                               |
| Triggers                                                  | When the customer clicks on the “Buy” button .                                                                                                                                                                                                           |  |       |        |                                                           |                                                                                                               |
| Pre-conditions                                            | i. User must be logged in.<br>ii. Car must be available.                                                                                                                                                                                                 |  |       |        |                                                           |                                                                                                               |
| Post-conditions                                           | System shows purchase confirmation.                                                                                                                                                                                                                      |  |       |        |                                                           |                                                                                                               |
| Basic Flow                                                | <table><tr><td>Actor</td><td>System</td></tr><tr><td>1. User clicks “Buy”.<br/>2. User fills form and confirms.</td><td>1. System verifies login and displays order form.<br/>2. System stores order and displays confirmation message</td></tr></table> |  | Actor | System | 1. User clicks “Buy”.<br>2. User fills form and confirms. | 1. System verifies login and displays order form.<br>2. System stores order and displays confirmation message |
| Actor                                                     | System                                                                                                                                                                                                                                                   |  |       |        |                                                           |                                                                                                               |
| 1. User clicks “Buy”.<br>2. User fills form and confirms. | 1. System verifies login and displays order form.<br>2. System stores order and displays confirmation message                                                                                                                                            |  |       |        |                                                           |                                                                                                               |
| Exceptions                                                | 1. User not logged in .<br>Solution: Redirect to Login page .                                                                                                                                                                                            |  |       |        |                                                           |                                                                                                               |
| Qualities                                                 | <ul style="list-style-type: none"><li>● Instant response .</li><li>● Offline functionality.</li></ul>                                                                                                                                                    |  |       |        |                                                           |                                                                                                               |

## 2.3 User Stories

### Admin

- As an Admin , I want to **log in** so that I can access my admin panel to manage cars.
- As an Admin, I want to **add a new car** so that I can update the car list with newly available cars.
- As an Admin, I want to **update car details** so that I can update the car list with newly available cars.
- As an Admin , I want to **view the list of cars** so that I can monitor available stock.
- As an Admin , I want to **delete a car** so that I can remove outdated or sold cars from the system.
- As an Admin , I want to **log out** from the system so that my admin account remains secure.

### Customer

- As a Customer, I want to **register an account** so that I can access the website and explore cars.
- As a Customer, I want to **log in** so that so that I can securely access my dashboard.
- As a Customer, I want to **view the list of available cars** so that I can choose one that matches my interest.
- As a Customer, I want to **view detailed information** of a selected car so that I can learn about its features and price from buying.
- As a Customer , I want to **buy a car** so that I can stimulate a car purchase and receive confirmation.
- As a customer ,I want to **logout** from the session so that I can securely end my session.

## 2.4 Storyboards

### Admin:

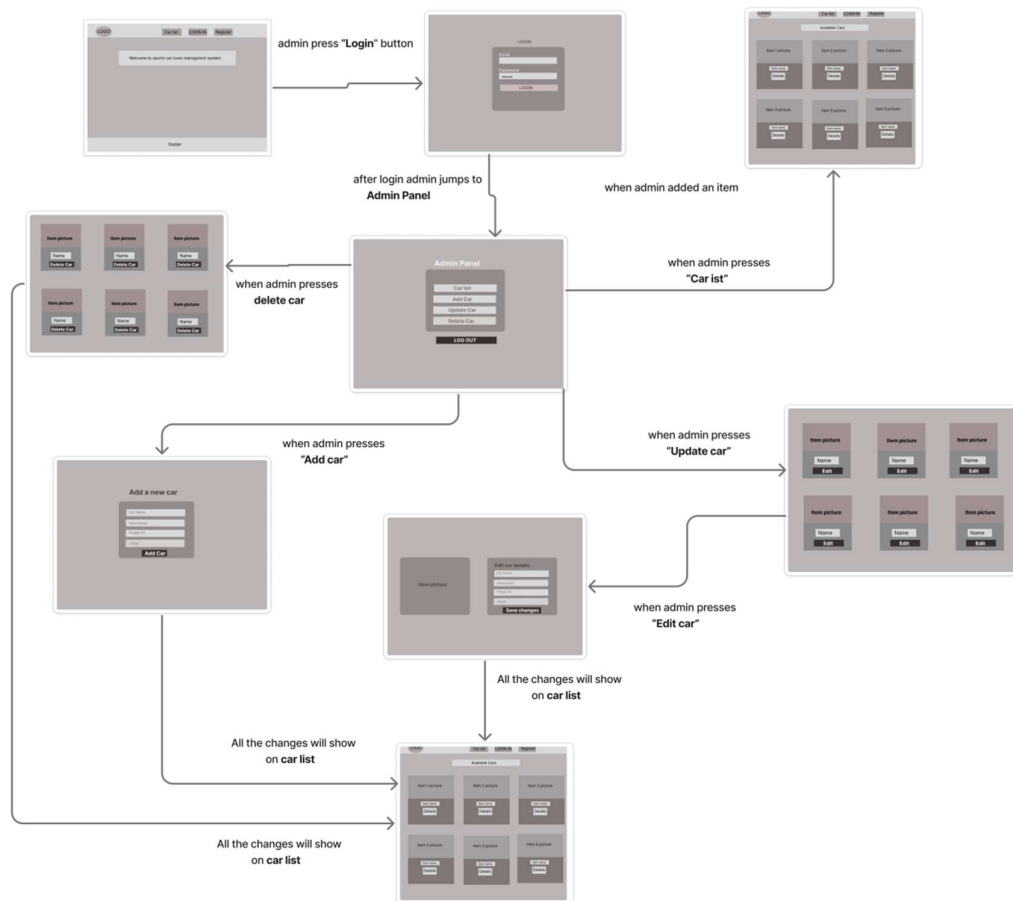


Figure 13: Story board as admin

## Customer:

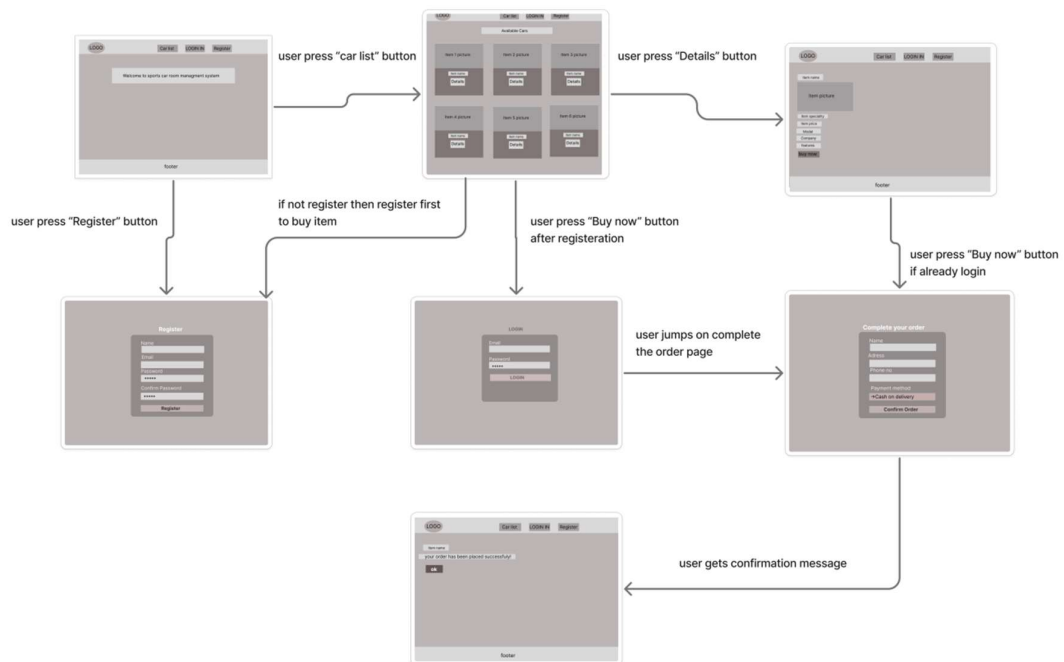


Figure 14 : Story board as customer

## References

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